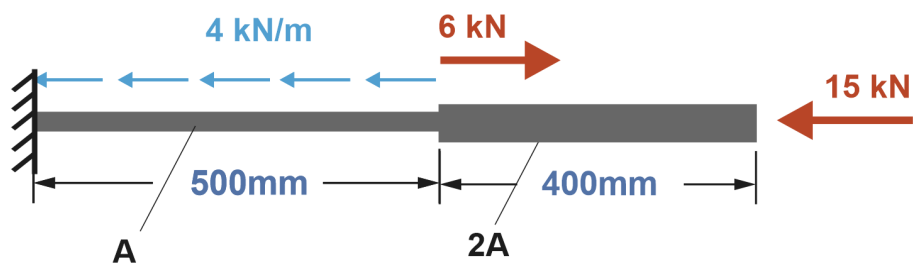


Analiza aksijalno opterećenog štapa metodom konačnih elemenata



Matrica krutosti elementa

$$k_{el} = \frac{EA}{l} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

```
clc; clear;  
E=210000; % MPa  
A=225; % mm^2  
p=-4; % N/mm
```

E- Youngov modul elastičnosti, A površina poprečnog presjeka

Tablica koordinata čvorova

broj čvora	x
1	0
2	500
3	900

Tablica povezanosti elemenata i čvorova

broj elementa	čvorovi	stupnjevi slobode
1	1, 2	1, 2
2	2, 3	2, 3

Slaganje globalne matrice krutosti

Ukupan broj stupnjeva slobode

```
n_dof = 3;
```

Inicijaliziranje globalne matrice krutosti, globalnog vektora sila u čvorovima i globalnog vektora kontinuiranog opterećenja

```
KG=zeros(n_dof,n_dof);  
FG=zeros(n_dof,1);
```

```
FG_p=zeros(n_dof,1);
```

Element 1

```
l = 500; % mm  
el_1_dofs = [1 2];  
  
k_el_1 = (E*A/l)*[1 -1; -1 1]
```

```
k_el_1 = 2x2  
    94500    -94500  
   -94500     94500
```

```
fp_el_1 = [p*l/2 p*l/2]'
```

```
fp_el_1 = 2x1  
   -1000  
   -1000
```

Dodavanje matrice krutosti elementa 1 globalnoj matrici krutosti

```
KG(el_1_dofs,el_1_dofs) = KG(el_1_dofs,el_1_dofs) + k_el_1
```

```
KG = 3x3  
    94500    -94500         0  
   -94500     94500         0  
         0         0         0
```

Dodavanje vektora kontinuiranog opterećenja

```
FG_p(el_1_dofs) = FG_p(el_1_dofs) + fp_el_1
```

```
FG_p = 3x1  
   -1000  
   -1000  
         0
```

Element 2

```
l = 400; % mm  
el_2_dofs = [2 3];  
  
k_el_2 = (E*2*A/l)*[1 -1; -1 1]
```

```
k_el_2 = 2x2  
    236250   -236250  
   -236250    236250
```

```
fp_el_2 = [0 0]'
```

```
fp_el_2 = 2x1  
         0  
         0
```

Dodavanje matrice krutosti elementa 2 globalnoj matrici krutosti

```
KG(el_2_dofs,el_2_dofs) = KG(el_2_dofs,el_2_dofs) + k_el_2
```

```
KG = 3x3
    94500    -94500         0
   -94500    330750   -236250
         0   -236250    236250
```

Dodavanje vektora kontinuiranog opterećenja

```
FG_p(e1_2_dofs) = FG_p(e1_2_dofs) + fp_e1_2
```

```
FG_p = 3x1
    -1000
    -1000
         0
```

Vektor koncentriranih sila

```
FG_c=zeros(n_dof,1)
```

```
FG_c = 3x1
         0
         0
         0
```

```
FG_c(2) = 6000; %N
FG_c(3) = -15000 %N
```

```
FG_c = 3x1
         0
        6000
       -15000
```

Vektor slobodnih pomaka (stupnjeva slobode)

```
free_dofs = [2 3]
```

```
free_dofs = 1x2
           2     3
```

Rješavanje sustava jednadžni

Inicijaliziranje globalnog vektora pomaka i vektora reakcija veza

```
UG = zeros(n_dof,1)
```

```
UG = 3x1
         0
         0
         0
```

```
RG = zeros(n_dof,1)
```

```
RG = 3x1
         0
         0
         0
```

Globalni vektor vanjskih sila

```
FG = FG_p + FG_c
```

```
FG = 3x1
      -1000
      5000
      -15000
```

Rješavanje reduciranog sustava jednažbi

```
KG(free_dofs,free_dofs)
```

```
ans = 2x2
      330750      -236250
      -236250      236250
```

```
FG(free_dofs)
```

```
ans = 2x1
      5000
      -15000
```

```
UG(free_dofs) = KG(free_dofs,free_dofs)\FG(free_dofs)
```

```
UG = 3x1
      0
      -0.1058
      -0.1693
```

Određivanje reakcija

```
RG = KG*UG-FG
```

```
RG = 3x1
      11000
      0
      0
```

Pomaci i unutrašnje sile u elementima

Element 1

```
l1 = 500;
A1=A;
p_el_1=-4;
el_1_dofs = [1 2];

u_1=UG(el_1_dofs(1));
u_2=UG(el_1_dofs(2));

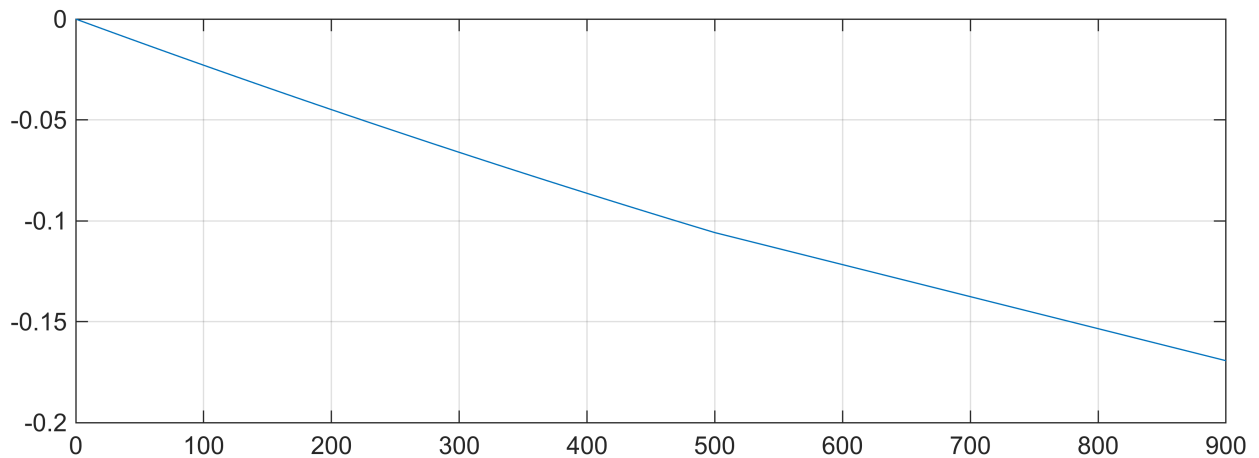
x=0:10:l1;
x1=0:10:l1;
u_el_1=(1-x/l1)*u_1 + (x/l1)*u_2 - (p_el_1/(2*E*A1))*(x.^2 -l1*x);
Nx_el_1=E*A1*(-1/l1*u_1 + 1/l1*u_2) - p_el_1*(x -l1/2);
```

Element 2

```
l2 = 400;  
A2 = 2*A;  
p_el_2 = 0;  
el_2_dofs = [2 3];  
  
u_1=UG(el_2_dofs(1));  
u_2=UG(el_2_dofs(2));  
  
x=0:10:l2;  
x2=l1:10:(l1+l2);  
u_el_2=(1-x/l2)*u_1 + (x/l2)*u_2 - (p_el_2/(2*E*A2))*(x.^2 -l2*x);  
Nx_el_2=E*A2*(-1/l2*u_1 + 1/l2*u_2) - p_el_2*(x-l2/2);
```

Crtanje dijagrama pomaka i unutrašnjih sila

```
set(gcf, 'Position', [100, 100, 1200, 400]); % 800x600 pixels  
plot([x1 x2],[u_el_1 u_el_2]);  
grid on
```



```
plot([x1 x2],[Nx_el_1 Nx_el_2]);  
ylim([-18000 0]);  
xlim([0 900]);  
grid on
```

